RAJALAKSHMI ENGINEERING COLLEGE

SUJITH S 251801341-2025-AIDS

M2

Started on	Thursday, 24 September 2026, 11:06 AM
State	Finished
Completed on	Thursday, 24 September 2026, 11:37 AM
Time taken	31 mins 4 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

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Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int             a;
    scanf("%d",&a);
    int x[a];
    int i;
    int             j;
    for(i=0;i<a;i++){
        scanf("%d ",&x[i]);
    }
    for(i=0;i<a;i++){
        for(j=1;j<a;j++){
            if(i!=j){
                if(x[i]==x[j]){
                    printf("%d",x[i]);
                    return 0;
            }}
        }
    }
}
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

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RAJALAKSHMI ENGINEERING COLLEGE

SUJITH S 251801341-2025-AIDS

M2

Started on	Saturday, 3 October 2026, 12:58 PM
State	Finished
Completed on	Saturday, 3 October 2026, 1:23 PM
Time taken	24 mins 43 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

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Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int slow = arr[0];
    int fast = arr[0];


    do {
        slow = arr[slow];
        fast = arr[arr[fast]];
    } while (slow != fast);


    slow = arr[0];

    while (slow != fast) {
        slow = arr[slow];
        fast = arr[fast];
    }

    printf("%d\n", slow);

    return 0;
}
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓

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	Input	Expected	Got	
	5 1 2 3 4 4	4	4	
	5 1 1 2 3 4	1	1	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

SUJITH S 251801341-2025-AIDS

M2

Started on	Thursday, 6 August 2026, 11:02 AM
State	Finished
Completed on	Thursday, 6 August 2026, 11:08 AM
Time taken	6 mins 26 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

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Question 1 | Correct Mark 1.00 out of 1.00

Converttheffollowing algorithm into a program and find its time complexity using counter method.

```
Factor(num){
{
   for (i = 1; i <= num;++i)
   {
    if(num%i==0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>
int count=0;
void Factor(int num){
    for(int i=1;i<=num;++i){
        count++;
        count++;
        if(num%i==0){
            count++;
        //    printf("%d ",i);
        }
    }
    count++;
}
int main(){
    int num;
    scanf("%d",&num);
    Factor(num);
    printf("%d",count);
}
```

	Input	Expected	Got	
				12	31	31				
	25	54	54	
	4	12	12	

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Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

SUJITH S 251801341-2025-AIDS

M2

Started on	Thursday, 6 August 2026, 11:08 AM
State	Finished
Completed on	Thursday, 6 August 2026, 11:12 AM
Time taken	3 mins 31 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

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Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
voidfunction(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>
int count=0;
void function(int n){
    int c=0;
    count++;
    for(int i=n/2;i<n;i++){
        count++;
        for(int j=1;j<n;j=2*j){
            count++;
            for(int k=1;k<n;k=k*2){
                count++;
                c++;
                count++;
            }
            count++;
        }
        count++;
    }
    count++;
}
int main(){

    int n;
    scanf("%d",&n);
    function(n);
    printf("%d",count);
}
```

	Input	Expected	Got	
✓	4	30	30	✓
	10	212	212	

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Passed all tests!

Correct

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RAJALAKSHMI ENGINEERING COLLEGE

SUJITH S 251801341-2025-AIDS

M2

Started on	Saturday, 3 October 2026, 2:01 PM
State	Finished
Completed on	Saturday, 3 October 2026, 2:06 PM
Time taken	5 mins 25 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

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Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    int k;
    scanf("%d", &k);

    int i = 0, j = 1;

    while (j < n) {
        int diff = a[j] - a[i];

        if (diff j &&      == k) {
            printf("1\n");
            return 0;
        }

        if (diff < k)
            j++;
        else
            i++;

        if (i == j)
            j++;
    }
```


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```
33
34      printf("0\n");
35
36      return 0;
37  }
```

	Input	Expected	Got	
	3 1 3 5 4	1	1	
	10 1 4 6 8 12 14 15 20 21 25 1	1	1	
	10 1 2 3 5 11 14 16 24 28 29 0	0	0	
	10 0 2 3 7 13 14 15 20 24 25 10	1	1	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

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